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Regression function

For square loss function $L(Y, f(X)) = (Y - f(X))^2$, the expected prediction error is

$$\text{EPE}(f) = \mathbb{E}[(Y - f(X))^2].$$

It is easy to see that the regression function $f(x) = \mathbb{E}[Y \mid X = x]$ minimizes the expected prediction error. Indeed, as all $\sigma(X)$ -measurable random variables are of the form $f(X)$, it is equivalent to minimize $\|Y - Z\|_2$ over all $\sigma(X)$ -measurable random variables Z . This is of course achieved by the orthogonal projection of Y onto the closed subspace of $\sigma(X)$ -measurable functions which is exactly $\mathbb{E}[Y \mid X]$.

The nearest neighbour method directly tries to estimate this regression function. In particular, intuitively, given iid $(X_i, Y_i)_{i=1}^n$ and x , we hope to approximate

$$\mathbb{E}[Y \mid X = x] \approx \frac{1}{k} \sum_{i=1}^n \mathbf{1}_{\{\|X_i - x\| \leq \|X_{(k)} - x\|\}} Y_i.$$

Namely, we take the sample average of the Y_i values corresponding to the k nearest neighbours of x . By taking a variance-bias decomposition, if we take k, n in such a way that $k \rightarrow \infty$ and $\frac{k}{n} \rightarrow 0$, then under some mild assumptions on the distribution of (X, Y) , we have the approximation converges in the limit.

The nearest neighbour method (and local methods in general) is problematic in high dimensions. For fixed x , if we want our estimate to depend on the data within a neighbourhood of x which captures a fixed proportion of the data, then the radius of this neighbourhood has to grow exponentially in the dimension (so rather contradicts the intuition of locality). This is called *the curse of dimensionality*.

A natural extension of the nearest neighbour method is, instead of taking the average of the nearest neighbours, we can take a weighted average of all points weighted by their distance to x . We call the weight function K the kernel function and a simple estimate (Nadaraya-Watson) is

$$f(x) = \frac{1}{Z} \sum_{i=1}^n K(x, X_i) Y_i$$

with the normalisation $Z = \sum_{i=1}^n K(x, X_i)$.

Linear regression

Let us consider the OLS estimator from a probabilistic perspective. Suppose $\mathcal{H}, \mathcal{K}$ are two Hilbert spaces and we have random variables $X \in \mathcal{H}$ and $Y \in \mathcal{K}$. The fundamental assumption one makes with linear regression is that the regression function is linear, i.e. there exists a linear operator $\beta : \mathcal{H} \rightarrow \mathcal{K}$ such that $\mathbb{E}[Y | X] = \beta X$ (we assume w.l.o.g. that both X, Y are centred).

Now, recalling that the covariance operators C_X and C_{XY} are defined by

$$C_X : \mathcal{H} \rightarrow \mathcal{H}, \quad C_X f = \mathbb{E}[\langle f, X \rangle_{\mathcal{H}} X]$$

and

$$C_{XY} : \mathcal{H} \rightarrow \mathcal{K}, \quad C_{XY} f = \mathbb{E}[\langle f, X \rangle_{\mathcal{H}} Y].$$

Thus, we compute

$$\begin{aligned} \langle C_{XY} h, k \rangle &= \mathbf{E}[\langle X, h \rangle \langle Y, k \rangle] = \mathbf{E}[\langle X, h \rangle \langle \mathbf{E}[Y | X], k \rangle] \\ &= \mathbf{E}[\langle X, h \rangle \langle \beta X, k \rangle] = \langle \beta C_X h, k \rangle \end{aligned}$$

for all $h \in \mathcal{H}$ and $k \in \mathcal{K}$. Hence, we have $C_{XY} = \beta C_X$. Thus, in the case where C_X is invertible, we have $\beta = C_{XY} C_X^{-1}$.

In the finite-dimensional case, the OLS estimator is exactly the empirical version of this formula. In particular, suppose now $\mathcal{H} = \mathbb{R}^p$ and $\mathcal{K} = \mathbb{R}$. Then, we have the covariance matrices $C_X = \mathbb{E}[XX^\top]$ and $C_{XY} = \mathbb{E}[XY^\top]$. Now, given iid data $(\mathbf{X}, \mathbf{Y}) = (X_i, Y_i)_{i=1}^n \in \mathbb{R}^{n \times p} \times \mathbb{R}^n$, we can estimate C_X and C_{XY} by their empirical versions

$$\hat{C}_X = \frac{1}{n} \sum_{i=1}^n X_i X_i^\top = \frac{1}{n} \mathbf{X}^\top \mathbf{X}, \quad \hat{C}_{XY} = \frac{1}{n} \sum_{i=1}^n X_i Y_i^\top = \frac{1}{n} \mathbf{X}^\top \mathbf{Y}.$$

Thus, the OLS estimator is exactly $\hat{\beta} = \hat{C}_{XY} \hat{C}_X^{-1}$. Note that the OLS estimator one typically writes down is the transpose of what we have here as one typically instead assumes $Y = X\beta + \varepsilon$.

The Cameron-Martin space and the RKHS

Suppose now X is a Gaussian random variable on some Banach space $\mathcal{B}$ with law μ . We recall that the Cameron-Martin space $\mathcal{H}_\mu$ of μ is defined as the closure (with respect to the Cameron-Martin norm) of the pre-Cameron-Martin space

$$\mathcal{H}_\mu^\circ = \{x \in \mathcal{B} : \exists x^* \in \mathcal{B}^*, l(x) = C_\mu(l, x^*) \forall l \in \mathcal{B}^*\} = C_\mu(\mathcal{B}^*)$$

where in the latter equality we identify $C_\mu : \mathcal{B}^* \rightarrow \mathcal{B}$ by $C_\mu l = \int l(x) x \mu(dx)$ with the integral understood in the Bochner sense.

We show that the Cameron-Martin space is exactly the RKHS (in the statistician's sense) associated to the kernel $k(h, k) = C_\mu(h, k) = \int h(x) k(x) \mu(dx)$ (to be more precise, the RKHS of k is the canonical identification of $\mathcal{H}_\mu$ in $\mathcal{B}^{**}$). Note that the RKHS here is different from the RKHS in [Haia, Haib] which is instead the $L^2(\mu)$ -closure of $\mathcal{B}^*$.

Recall that the RKHS $\mathcal{H}$ associated to a reproducing kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is a Hilbert space of functions $f : \mathcal{X} \rightarrow \mathbb{R}$ such that $k(x, \cdot) =: k_x \in \mathcal{H}$ for all x and

$$\langle f, k(x, \cdot) \rangle_{\mathcal{H}} = f(x).$$

In general, the RKHS $\mathcal{H}$ is canonically defined by $\mathcal{H} = \overline{\mathcal{H}^\circ} = \overline{\langle k(x, \cdot) : x \in \mathcal{X} \rangle}^{\mathcal{H}}$ with the closure taken with respect to the topology generated by the inner product

$$\left\langle \sum_i \alpha_i k(x_i, \cdot), \sum_j \beta_j k(x_j, \cdot) \right\rangle_{\mathcal{H}} = \sum_{i,j} \alpha_i \beta_j k(x_i, x_j).$$

In this case, $\mathcal{X} = \mathcal{B}^*$ and $\mathcal{H} = \mathcal{H}_\mu$ with the canonical identification $x \in \mathcal{H}_\mu \subseteq \mathcal{B}^{**}$, $x : l \mapsto l(x)$. Indeed, in this case, $k(l, \cdot) = C_\mu l \in \mathcal{H}_\mu^\circ$ and for any $x \in \mathcal{H}_\mu^\circ$, we have that

$$\langle x, k(l, \cdot) \rangle_\mu = \langle x, C_\mu l \rangle_\mu = C_\mu(x^*, l) = l(x) = x(l)$$

where we used the fact that $(C_\mu l)^* = l$.

Conversely, suppose we have the reproducing kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ (recall it is symmetric and positive definite) and its associated RKHS $\mathcal{H}$. In this case, we consider a isonormal Gaussian process W on $\mathcal{H}$ (and its first chaos $\mathcal{H}_1$ is isomorphic to $\mathcal{H}$ via the map $h \mapsto W(h)$). On the other hand, under some mild regularity assumptions (e.g. via Kolmogorov continuity criterion), there exists a Gaussian measure μ on some Banach space $\mathcal{B} \subseteq \mathbb{R}^{\mathcal{H}}$ for which the Cameron-Martin space is exactly $\mathcal{H}_1$. Thus, in this case, the above correspondence between the Cameron-Martin space and the RKHS holds as well.

With this language in mind, viewing X as an isonormal Gaussian process over an RKHS $\mathcal{H}$, linear regression corresponds to finding the component of the regression function in the first chaos of X . Namely, taking $Z_n \in \mathcal{H}_n$ with $Z_1 = X(h)$ such that

$$\mathbf{E}[Y | X] = \sum_{n=1}^{\infty} Z_n =: Z \in L^2 = \bigoplus_{n=1}^{\infty} \mathcal{H}_n,$$

linear regression corresponds to estimating $h \in \mathcal{H}$ (or equivalently, estimating DZ - the Malliavin derivative of Z against X).

Regression and kernels

For practising statisticians, the RKHS does not arise from some abstract Gaussian measure but instead, is chosen a priori to capture some belief on the regression function. In particular, the workflow is to choose the RKHS $\mathcal{H}$ from which we assume $f \in \mathcal{H}$ and it solving a minimisation problem

$$f = \arg \min_{f \in \mathcal{H}} c(y, x, f(x)) + J(\|f\|_{\mathcal{H}}^2) \quad (0.1) \quad \boxed{\text{eq:regression-kernel}}$$

for some $x \in \mathcal{X}^n, y \in \mathbb{R}^n$ data points, some cost function c and some regularisation function J . From this, the kernel arises naturally via the representer theorem.

Theorem 0.1 (Representer theorem [Sha]) Let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be the reproducing kernel of $\mathcal{H}$. Then, $\hat{f} \in \mathcal{H}$ minimises (0.1) if and only if $\hat{f}$ is of the form $\hat{f} = \sum_{i=1}^n \alpha_i k(x_i, \cdot)$ for some $\alpha = (\alpha_i)_{i=1}^n \in \mathbb{R}^n$ minimising

$$\arg \min_{\alpha \in \mathbb{R}^n} c(y, x, K\alpha) + J(\alpha^\top K\alpha)$$

where $K \in \mathbb{R}^{n \times n}$ with $K_{ij} = k(x_i, x_j)$.

In the case of the OLS and ridge regression, the RKHS is simply $\mathcal{H} = \mathbb{R}^p$ with the linear kernel $k(x, y) = x^\top y$ in which one minimizes

$$\hat{\beta} = \arg \min_{f \in \mathcal{H}} \left\{ \sum_{i=1}^n (Y_i - f(X_i))^2 + \lambda \|f\|_{\mathcal{H}}^2 \right\}$$

for some regularisation parameter $\lambda \geq 0$ (with $\lambda = 0$ corresponding to OLS).

We note the existence of a eigenfunction decomposition of the kernel k (Mercer's theorem) under mild assumptions. In particular, we assume k a reproducing kernel and $X \sim \mu \in \mathcal{M}(\mathcal{B}^*)$ (NB. the covariance of X here is NOT the kernel k and in fact we do not assume X to be Gaussian) such that $k \in L^2(\mu \otimes \mu)$. Then, for any $e \in \mathcal{H}$ - the RKHS associated to k , we say e is an eigenfunction of k if there exists $\lambda \in \mathbb{R}$ such that for any $x \in \mathcal{X}$,

$$\mathbb{E}[k(x, X)e(X)] = \lambda e(x) \quad (0.2) \quad \text{eq:eigenfunction-ke}$$

or equivalently (by recalling the “continuous evaluation” property of the RKHS),

$$\langle k_x, e \rangle_{L^2(\mu)} = \lambda \langle k_x, e \rangle_{\mathcal{H}}.$$

Then, Mercer's theorem states the following.

Theorem 0.2 (Mercer's theorem [Sha]) Under the above setting with some additional mild conditions, we have the decomposition

$$k(x, y) = \sum_{i=1}^{\infty} \lambda_i e_i(x) e_i(y)$$

for some $(e_i)_{i=1}^{\infty}$ an orthonormal (in $L^2(\mu)$) eigenbasis of k with corresponding eigenvalues $(\lambda_i)_{i=1}^{\infty}$ with the sum counting multiplicities.

Let us discuss how the definition via (0.2) can be viewed as a bona fide eigenfunction of an operator on $\mathcal{H}$. Consider the canonical embedding $\iota : \mathcal{H} \hookrightarrow L^2(\mu)$ and its adjoint $\iota^* : L^2(\mu) \rightarrow \mathcal{H}$. For any $h \in \mathcal{H}$, we have that

$$\iota^* \iota(h) = (l \in \mathcal{H} \mapsto \langle \iota(h), \iota(l) \rangle_{L^2(\mu)})$$

where the right hand side is identified with an element in $\mathcal{H}$ via the Riesz representation theorem. We claim that $e \in \mathcal{H}$ is a eigenfunction of k if and only if it is an eigenfunction of $\iota^* \iota$. Indeed, if $\lambda e \equiv \lambda \langle e, \cdot \rangle_{\mathcal{H}} = \iota^* \iota(e)$, we have that for any $x \in \mathcal{X}$,

$$\langle k_x, e \rangle_{L^2(\mu)} = \iota^* \iota(e) k_x = \lambda \langle e, k_x \rangle_{\mathcal{H}} = \lambda e(x)$$

Conversely, if e is an eigenfunction of k , we have that

$$\iota^* \iota(e) k_x = \langle e, k_x \rangle_{L^2(\mu)} = \lambda \langle e, k_x \rangle_{\mathcal{H}}$$

Thus, as $\mathcal{H}$ is the linear span of k_x , we have $\iota^* \iota(e) = \lambda e$ as well.

With this in mind, Mercer's theorem can be viewed as a corollary of the spectral theorem under some additional assumptions (e.g. compactness of $\iota^* \iota$). In particular, assuming we have a orthonormal (in $L^2(\mu)$) eigenbasis $(e_i)_{i=1}^\infty$ of $\iota^* \iota$ in $\mathcal{H}$, we have that for any $x, y \in \mathcal{X}$,

$$k(x, y) = \left\langle k_x, \sum_{i=1}^{\infty} \langle k_y, e_i \rangle_{L^2(\mu)} e_i \right\rangle_{\mathcal{H}} = \sum_{i=1}^{\infty} \iota^* \iota(e_i) k_y \langle k_x, e_i \rangle_{\mathcal{H}} = \sum_{i=1}^{\infty} \lambda_i e_i(x) e_i(y)$$

as required.

References

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